

Mid-West University
Examinations Management Office

End-Semester Examinations -2080

Bachelor level/ B.E. Civil/3rd Semester

Time: 3 hours

Full Marks: 50

Pass Marks: 25

Subject: Building Construction and Building Drawing (CE435/CE208)

- Attempt all questions
- Figures in the margin indicate full marks.
- Assume suitable values, with a stipulation, if necessary.
- Candidates are required to answer the questions in their own words as far as possible.

1. a) What do you mean by orientation of building? Briefly discuss the factors to be considered for best orientation of a building? [6]
b) What is damp proofing? What are the cause and effect of dampness? [4]
2. a) What are the requirement of foundation? Explain the types of foundation with neat sketches. [6]
b) What is bond? Describe the different types of brick Masonry. [4]
3. a) Define Roof Covering. Explain commonly used roof covering for pitched roof. [4]
b) Design an Open well staircase of public building which measures the 6m x 7m. The height of the floor is 4.5m [6]
4. a) Explain the types of windows and their fixing details of windows. [4]
b) Briefly explain the detailing the joint for roof and floor level. [4]
5. Write the short notes (Any Four) [4x3=12]
 - a) Scaffolding and its types
 - b) Cladding
 - c) Rainwater Harvesting
 - d) Retaining Walls
 - e) Ventilation & Air Conditioning

THE END

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Examinations Management Office
End-Semester Examinations -2080

Bachelor level/ B.E. Civil/3rd Semester

Time: 3 hours

Subject: Communication English (SH432/SH207)

Full Marks: 50

Pass Marks: 25

- Attempt all the questions
- Figures in the margin indicate full marks.
- Assume suitable values, with a stipulation, if necessary.
- Candidates are required to answer the questions in their own words as far as possible.

Group A

[6x5=30]

Short Questions

1. What are the fundamentals of effective speaking? Describe.
2. What activities do you recommend for notice preparation? Describe.
3. What is reading? Describe it with its types.

OR

Write down the format of technical report.

4. Describe the procedure of writing a research article.
5. What do you mean by reference? Define with examples.
6. What are the stages of technical writing process? Describe.

OR

What is proposal? Describe any five key components.

Group B

[2x10=20]

Long Questions

7. Write an application letter for the post of Civil Engineer stating suitable qualifications and experiences.

8. **Read the following passage and answer the questions given below:**

Daniel Defoe was born in London in 1660. He was not very educated and spent many years of his young life studying religion and later traveling as a merchant. He traveled widely and built up a successful business. During this period, he married and started raising a family. However, around 1692, his business failed and he fell into debt. Since he had, always been interested in Politics, he tried to earn money by writing Political articles for the newspapers. But his political writing brought only troubles and increasing debts, and so Defoe turned to fiction writing. His first novel, written in 1719, when Defoe was nearly sixty years old, was to become one of the best-known adventure stories in the world. The novel was Robinson Crusoe-a story which thrills readers even today, more than two hundred and fifty years later. Robinson Crusoe brought Defoe great success and helped him pay back part of his debts. He continued writing novels such as Moll flinders, Colonel jack, and two other Robinson Crusoe's stories, but none became as popular as the first one.

Questions:

- a) Who was Daniel Defoe?
- b) Why did Defoe give up writing political articles?
- c) What was Defoe's best known novel?
- d) What are the novels that Daniel wrote?
- e) How did he clear his debts?

OR

Highlight the importance of Communication English in the field of engineering giving convincing examples and reasons.

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End-Semester Examinations -2080

Bachelor level/ B.E. Civil/3rd Semester

Time: 3 hours

Subject: Engineering Geology (CE434/CE207)

Full Marks: 50

Pass Marks: 25

- Attempt all questions
- Figures in the margin indicate full marks.
- Assume suitable values, with a stipulation, if necessary.
- Candidates are required to answer the questions in their own words as far as possible.

1. Describe scope and objectives of geology in the field of civil engineering? [5]
2. Suppose you are hired to survey 10Km road at Chankheli Rural Municipality, Humla. What types of geological data will you collect? Also write about the importance of geology in infrastructure construction. [5]
3. What are the components of the Earth? Describe internal structure of the Earth with a diagram. [5]
4. Define petrology. How do you identify igneous rocks in the field? Define civil engineering significance of Phyllite, Granite and Sandstone. [10]
5. a) Distinguish between erosion and weathering. [2.5]
b) How would you carry geological work in running water? [2.5]
6. What is geological structure? How folds are formed? [5]
7. Discuss major discontinuities system of the Nepal Himalaya with their engineering significance. [5]
8. Describe the tectonic division of Nepal. Explain geology of Lesser Himalaya. [5]
9. Write short notes on: [5]
 - a) P and S wave
 - b) Plate Boundaries

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Mid-West University
Examinations Management Office
 End-Semester Examinations -2080

Bachelor level/ B.E. Civil/3rd Semester

Time: 3 hours

Subject: Surveying I (CE433/CE206)

Full Marks: 50

Pass Marks: 25

- Attempt all questions
- Figures in the margin indicate full marks.
- Assume suitable values, with a stipulation, if necessary.
- Candidates are required to answer the questions in their own words as far as possible.

1. Differentiate between topographical survey and cadastral survey. Draw a conventional symbol of bridge and benchmark. [2]
2. Define and differentiate between error and mistake. Explain the concept of precision. [2+1]
3. A base line was measured by tape suspended in catenary under a pull of 145 N, the mean temperature being 14° C. The length of various segments of the tape and the difference in level of the two ends of a segment are given below:

Bay	Length (m)	Difference in Level (m)
1	29.988	+0.346
2	29.895	-0.214
3	29.838	+0.309
4	29.910	-0.106

If the tape was standardized on the flat under a pull of 95 N at 18 ° C. Determine the correct length of the line. Take cross sectional area of the tape = 3.35 sq.mm

Mass of the tape= 0.025 kg/m

Coefficient of linear expansion = 0.9×10^{-6} per °C

Young's Modulus= 14.8×10^4 MN/ sq.m

Mean height of the line above M.S.L.= 51.76 m

Radius of the earth= 6370 km.

4. What is deflection angle? The following observations were recorded in a theodolite traverse ABCDA. Compute the mean horizontal angles and adjust them if necessary. [4]

Inst. Station	Target Station	HCR observation	
		Face Left	Face Right
A	D	90°00'00"	269°59'30"
	B	204°25'40'	24°25'30'
B	A	90°00'00"	270°00'30"
	C	190°36'10'	10°36'00'
C	B	90°00'00"	269°59'50"
	D	169°08'40'	349°09'20'
D	C	90°00'00"	270°00'00"
	A	165°12'40'	345°12'30'

5. Explain the obstacles in chaining and ranging with figure. The following angles were observed in the clockwise direction in an open traverse.

$$\angle ABC = 124^\circ 15'$$

$$\angle BCD = 156^\circ 30'$$

$$\angle CDE = 102^\circ 00'$$

$$\angle DEF = 95^\circ 15'$$

$$\angle EFG = 215^\circ 45'$$

The magnetic bearing of the line AB was $242^\circ 34'$. What would be the bearing of the line FG. [2+4]

[3]

6. Balance the following traverse by Bowditch's rule.

S.No	Line	Length(m)	Latitude	Departure
1	AB	200	-173.20	+100.0
2	BC	130	00.0	+130.0
3	CD	100	+86.6	+50.0
4	DE	250	+250.0	0.0
5	EA	320	-154.90	-280

7. The following observations were made for a closed traverse round an obstacle. Due to obstructions, bearings of lines BC and CD could not be measured. Find out the missing bearings. [4]

Line	Length(m)	Bearing
AB	550	60°
BC	1200	?
CD	880	?
DA	1050	310°

8. During a fly leveling operation, the following observations were made:

BS- 1.315, 2.035, 1.980, 2.625 m

FS- 1.150, 3.450, 2.255 m

The first back sight was taken on a BM of RL 250.000 m. From the last position of instrument, it is required to set five pegs each at a distance of 20 m on a uniform rising gradient of 1 in 40. The first peg is to have RL of 247.245 m. Work out the staff readings required for setting the tops of the pegs on the given gradient. [5]

9. The reduced level of a factory floor is 30.00 m and the staff reading on the floor is 1.40 m. The staff reading when held inverted with the bottom touching the Tee-beam of the roof is 3.67 m. Find the height of the beam above the floor. [3]

10. Compare the radiation and intersection methods of plane table surveying and give an example of the most desirable application of each. [2]

11. What is meant by face left and face right of a theodolite? How would you change face? What instrumental errors are eliminated by face right and face left observations. [1+1+1]

12. Explain classification of triangulation system. Write Principle of triangulation and trilateration. [3+2]

13. State prismoidal formula for the calculation of volume. Calculate using Simpson's rule the area enclosed between the boundaries of the field, offsets to which have been taken from a chain line at intervals of 25 m to the right and left. [1+4]

Offset right (m)	31.9	27.5	48.9	56.6	18.3	25.4	41.6
Distance(m)	0	25	50	75	100	125	150
Offset left(m)	33.6	29.8	55.3	47.2	24.6	21.7	18.1

THE END

Mid-West University
Examinations Management Office
 End-Semester Examinations -2080

Bachelor level/ B.E. Civil/3rd Semester

Time: 3 hours

Subject: Fluid Mechanics (CE432/CE205)

Full Marks: 50

Pass Marks: 25

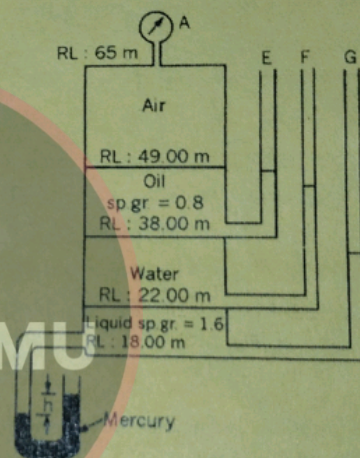
- Attempt all questions
- Figures in the margin indicate full marks.
- Assume suitable values, with a stipulation, if necessary.
- Candidates are required to answer the questions in their own words as far as possible.

1. Why does the viscosity of a liquid decreases with increase in temperature whereas it increases with increase in temperature in the case of a gas? Two large plane surfaces are 30 mm apart. This space is filled with glycerin of dynamic viscosity 0.82 Ns/m^2 . Find what force is required to drag a very thin plate of area 0.60 m^2 between the two surfaces at a speed of 0.70 m/s . [2+3]

- i) If the plate is equidistant from the two surfaces,
- ii) If the plate is 12 mm from one of the surfaces.

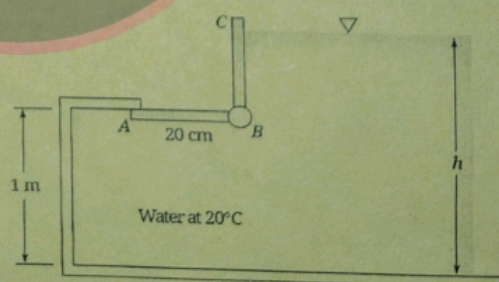
2. How do you find hydrostatics force on curve surface? In the given figure, R.L. of lowest level of Hg (i.e., at left limb) is 12 m. For gauge reading at A of 140 mm of mercury vacuum, determine:

- i) the elevations of the liquids in the open tubes E, F and G.
- ii) the value of h in the U-tube manometer.



[1+4]

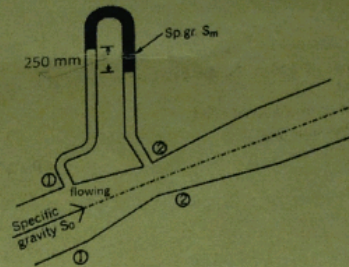
3. Gate ABC in figure has a fixed hinge line at B and is 2.5 m wide into the paper. The gate will open at A to release water if water depth is high enough. Compute the depth h for which the gate will begin to open. Neglect self-weight of the gate.



[4]

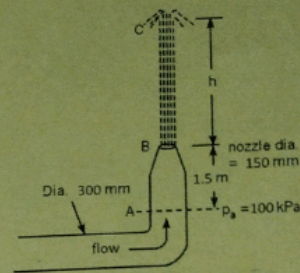
4. How would you determine the metacentric height experimentally? A cylindrical tank 2 m in diameter and 4 m in height contains water to a depth of 3 m. Find the speed of the tank so that 25 % of the original volume is spilled out. [2+3]
5. A stream function is given by, $\psi = 3x^2 - 3y^2$. Determine the velocity potential function. Derive [2+4]
 the continuity equation for three dimensional flow and hence reduced it for two dimensional steady incompressible flow using Cylindrical polar co-ordinate system.

6. Find the discharge of water flowing through a pipe of 350 mm diameter placed in an inclined position where a venturimeter is inserted, having a throat diameter of 175 mm. Find also the difference of pressure between the main and the throat which is measured by a liquid of specific gravity of 0.75 in an inverted U-tube which gives a reading of 250 mm. The loss of head between the main and the throat is 0.4 times the kinetic head of the liquid in the pipe.



[4]

7. Derive an expression for discharge over a right angled V-notch assuming coefficient of discharge as 0.60. Find the height h to which the water jet will shoot out of the nozzle in the arrangement shown in figure. Neglect loss of energy.



[4+3]

8. An orifice of 160 mm in diameter is fitted to the bottom of a cylindrical tank 3.5 m in diameter and 7 m high. A constant discharge of 100 litres of water per second is supplied into the tank and at the same time water is discharged through the orifice. Find the time taken to lower the water level in the tank from 5.5 m to 2.5 m above the bottom of the tank. The coefficient of discharge is 0.68. The top of the tank is open to the atmosphere. [4]
9. What are the various applications of impulse-momentum principle? The diameter of bend is 400 mm at inlet and 200 mm at outlet and the flow is turned through 120° in vertical plane, the axis of inlet is horizontal and the center of the outlet section is 1.6 m below the center of the inlet section, the total volume of fluid contained in the bend is 0.10 m^3 . Neglecting friction loss, Calculate the magnitude and direction of the force exerted by the water on the bend by the water flowing through it at 375 lps when the inlet pressure is 150 Kpa. [1+4]
10. Why distortion in model is necessary? The resisting force F of a supersonic plane depends upon the velocity of the aircraft V , its length L , air viscosity μ , air density ρ and the bulk modulus of air K . Show that the resisting force is a function of Reynolds number and Mach number. [1+4]

THE END

Mid-West University
Examinations Management Office
 End-Semester Examinations -2080

Bachelor level/ B.E. Civil/3rd Semester

Time: 3 hours

Subject: Engineering Mathematics III (SH431/SH203)

Full Marks: 50
 Pass Marks: 25

- Attempt all questions
- Figures in the margin indicate full marks.
- Assume suitable values, with a stipulation, if necessary.
- Candidates are required to answer the questions in their own words as far as possible.

1. a. (i) Find the linearly dependent or independent in the given vectors, $(1,2,-1), (2,3,0), (0,0,0)$.
 (ii) Find the Eigen value and Eigen vector of the following matrix. $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$. [2+3]

b. Proved that: $\begin{vmatrix} a^3 & 3a^2 & 3a & 1 \\ a^2 & a^2+2a & 2a+1 & 1 \\ a & 2a+1 & a+2 & 1 \\ 1 & 3 & 3 & 1 \end{vmatrix} = (a-1)^4(1+2a-2a^2)$. [5]

2. a. (i) Proved that the matrix $\begin{bmatrix} 1 & 1-i & 2 \\ 1+i & 3 & i \\ 2 & -i & 0 \end{bmatrix}$ is Hermitian. [2+3]

(ii) Find the Fourier sine series for the function $f(x) = e^{ax}$ $0 < x < \pi$.

b. (i) Find the Laplace transform of $t^2 e^{2t} \sin t$.

(ii) Define the inverse Laplace transform. Find the inverse transform by using Convolution theorem

$\frac{s^2}{(s^2+4)^2}$. [2+3]

3. a. Solve the following differential equation by Laplace transform method.
 $y''' + 2y'' - y' - 2y = 2e^{3t}$, given that $y(0) = 0, y'(0) = 0, y''(0) = 0$. [5]

b. (i) Show that $\vec{F} = yx\vec{i} + zx\vec{j} + xy\vec{k}$ is irrotational and find its scalar Potential function. [2+3]

(ii) Obtain the Fourier series for $f(x) = x^3$ in the interval $-\pi \leq x \leq \pi$.

4. a. State and proved Stokes theorem. [5]

b. Verify Greens theorem for $\oint_C [(2xy - x^2)dx + (x + y^2)dy]$. Where C is the closed curve of the region bounded by $y = x^2$ and $x = y^2$. [5]

5. a. Evaluate $\iint_S \vec{F} \cdot \hat{n} \, ds$, where $\vec{F} = xy\vec{i} + z^2\vec{j} + 2yz\vec{k}$ and S is the surface bounded by $x = 0, y = 0, z = 0$, and the plane $x + y + z = 1$ by Gauss divergence theorem. [5]

b. Define the Line integral. Evaluate $\int_C \vec{F} \cdot d\vec{r}$, for $\vec{F} = (2x - y)\vec{i} - yz^2\vec{j} - y^2z\vec{k}$ where S is the upper part of the sphere $x^2 + y^2 + z^2 = a^2$ and C is boundary. [5]

THE END

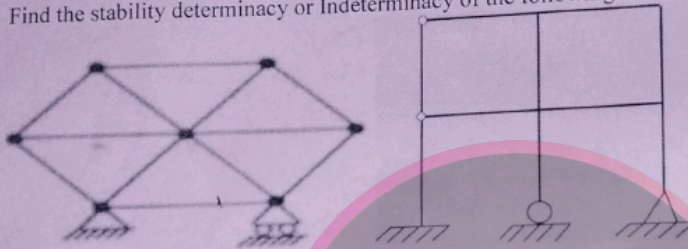
Mid-West University
Examinations Management Office
Final Examinations -2080

Bachelor level/ B.E. Civil/3rd Semester
Time: 3 hours
Subject: Strength of Materials (CE431/CE204)

Full Marks: 50
Pass Marks: 25

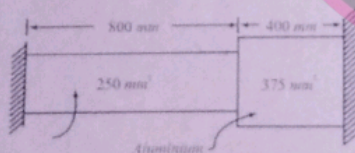
- Attempt all questions
- Figures in the margin indicate full marks.
- Assume suitable values, with a stipulation, if necessary.
- Candidates are required to answer the questions in their own words as far as possible.

1. a) Find the stability determinacy or Indeterminacy of the following trusses and Frames. [4]

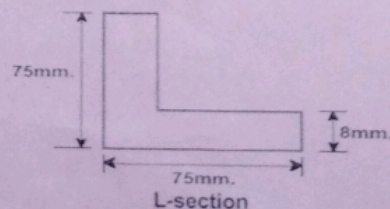


- b) A composite bar made up of aluminum and steel is held between two supports. The bars are stress free at 69°C . What will be the stresses in the bars when the temp. Drops to 49°C if (i) The supports are unyielding (ii) The supports come nearer to each other by 0.25 mm. [6]

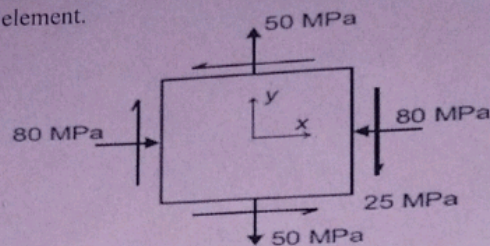
Take $E_s = 2 \times 10^5$ and $E_a = 0.7 \times 10^5 \text{ N/mm}^2$, $\alpha_s = 1.17 \times 10^{-5}$ and $\alpha_a = 2.34 \times 10^{-5}$ per $^{\circ}\text{C}$. The areas of steel and aluminium bars are 250 mm^2 and 375 mm^2 respectively.



2. a) Derive the relationship between the Elastic constants, i. e. E , ν and G . The symbols have their useful meanings. [4]
- b) Define Principal Axis? For the section shown in below figure determine the principal moment of inertia and Locate the position of principal Axis. [6]

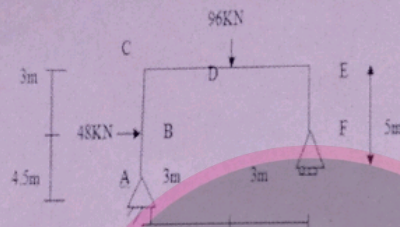


3. a) What is principal stress and principle angle? The given state of plane stress at a point is represented by the stress element below. Determine the principal stress and draw the corresponding stress element. [6]



- b) Derive the relation between Load, Shear force and Bending Moment of Beam. [4]

4. a) Determine the support reactions & Draw Axial force, Shear Force and Bending Moment Diagram for following given frame. Joint C and E are rigid joints. [6]

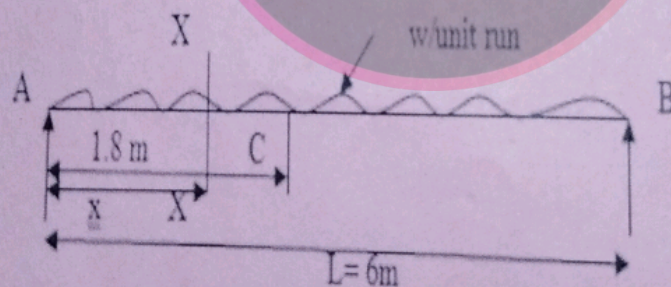


- b) Write the basic Assumptions in the theory of bending and derive the flexural equations. [4]

5. a) Define longitudinal, hoop and radial stress developed in thin cylinder subjected to internal pressure. [3]

- b) Derive the Euler's formula for critical load for column with both ends of column are fixed. [4]

- c) Calculate bending stress for a simply supported beam 6 m in span. The width of beam is 350 mm or depth of beam is 600 mm. The beam carries a UDL of 79kN/m over the whole span. [3]



THE END